Assalam-o-Alikum

Welcome to ShamsherAli Tech

AI Automation Full course

Lecture # 4

Python Basic to Advanced

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Strings

String is data type that stores a sequence of characters.

Basic Operations

- **concatenation**

  "hello" + "world" ⟶ "helloworld"

- **length of str**

  len(str)

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Indexing

A	p	n	a	_	C	o	l	l	e	g	e
0	1	2	3	4	5	6	7	8	9	10	11

str = "Apna_College"

str[0] is 'A', str[1] is 'p' ...

str[0] = 'B' #not allowed

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Slicing

Accessing parts of a string

```
str[ starting_idx : ending_idx ] #ending idx is not included

str = "ApnaCollege"

str[ 1 : 4 ] is "pna"

str[  : 4 ] is same as str[ 0 : 4]

str[ 1 :  ] is same as str[ 1 : len(str) ]
```

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Slicing

Negative Index

A	p	p	l	e
-5	-4	-3	-2	-1

str = "Apple"

str[-3 : -1] is "pl"

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String Functions

```
str = "I am a coder."

str.endsWith("er.")  #returns true if string ends with substr

str.capitalize( )  #capitalizes 1st char

str.replace( old, new )  #replaces all occurrences of old with new

str.find( word )  #returns 1st index of 1st occurrence

str.count("am")  #counts the occurrence of substr in string
```

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Let‘s Practice

WAP to input user’s first name & print its length.

WAP to find the occurrence of ‘$’ in a String.

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Conditional Statements

if-elif-else (SYNTAX)

```
if(condition) :
    Statement1
elif(condition):
    Statement2
else:
    StatementN
```

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Conditional Statements

Grade students based on marks

marks >= 90, grade = “A”

90 > marks >= 80, grade = “B”

80 > marks >= 70, grade = “C”

70 > marks, grade = “D”

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Let's Practice

WAP to check if a number entered by the user is odd or even.

WAP to find the greatest of 3 numbers entered by the user.

WAP to check if a number is a multiple of 7 or not.

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